

Hidden Markov Models

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Outline

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- **Objective:** Provide an intuitive introduction to Hidden Markov Models (HMMs), with a focus on their application in the Rayleigh fading channel and change point detection, important problems in digital communication.
- **Application domains:**
 - Speech recognition
 - Medical imaging and computational biology
 - Finance
 - Computer vision
 - Digital communications

Markov Chain

“A Markov chain is a stochastic process for which the future evolution only depends on the past through the current state.”

Definition and Theorem

Definition

Let $(X_n)_{n \geq 0}$ be a sequence of random variables taking values in I . We say that (X_n) is a **Markov Chain** with initial distribution λ and transition matrix P if for all $n \geq 0$ and all $i_0, i_1, \dots, i_{n+1} \in I$:

(a) $\mathbb{P}(X_0 = i_0) = \lambda_{i_0}$,

(b) $\mathbb{P}(X_{n+1} = i_{n+1} \mid X_0 = i_0, \dots, X_n = i_n) = P_{i_n, i_{n+1}}$.

We say that (X_n) is an $MC(\lambda, P)$ in short.

Theorem

A discrete-time random process $(X_n)_{n \geq 0}$ is $MC(\lambda, P)$ if and only if for all $n \geq 0$ and all $i_0, \dots, i_n \in I$,

$$\mathbb{P}(X_0 = i_0, \dots, X_n = i_n) = \lambda_{i_0} P_{i_0 i_1} \cdots P_{i_{n-1} i_n}.$$

Hidden Markov Models (HMM)

"A Hidden Markov model (HMM) is a Markov chain observed in noise."

- Indeed, the model comprises a Markov chain, which we will denote by $\{X_k\}_{k \geq 0}$, where k is an integer index.
- Now, the Markov chain is hidden. What is available to the observer is another stochastic process $\{Y_k\}_{k \geq 0}$, linked to the Markov chain.
- X_k governs the distribution of the corresponding Y_k .

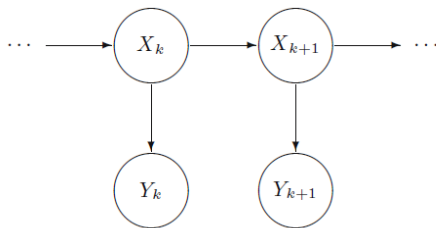


Figure: Graphical representation of the dependence structure of a hidden Markov model, where $\{Y_k\}$ is the observable process and $\{X_k\}$ is the hidden chain.

Definition

A hidden Markov model is a bivariate discrete-time process $\{(X_k, Y_k)\}_{k \geq 0}$, where $\{X_k\}$ is a Markov chain and, conditional on $\{X_k\}$, $\{Y_k\}$ is a sequence of independent random variables such that the conditional distribution of Y_k only depends on X_k .

- We will denote the state space of the Markov chain $\{X_k\}$ by X and the set in which $\{Y_k\}$ takes its values by Y .

We also now assume that each HMM is homogeneous, by which we mean that the Markov chain $\{X_k\}$ is homogeneous :its transition kernel does not depend on the time index, and that the conditional law of $\{Y_k\}$ does not depend on k either.

Gaussian Linear State-Space Models

Standard state-space model :

$$X_{k+1} = AX_k + RU_k$$

Hidden state

$$Y_k = BX_k + SV_k$$

Observation model

- $\{U_k\}_{k \geq 0}$: state or process noise
- $\{V_k\}_{k \geq 0}$: measurement noise

Both are independent standard (multivariate) Gaussian white noise sequences (i.i.d. Gaussian vectors with zero mean and identity covariance matrices);

- The initial condition X_0 is Gaussian with mean μ_X and covariance Γ_X , and is uncorrelated with the processes $\{U_k\}$ and $\{V_k\}$;
- All of the following matrices are known with appropriate dimensions.
 - A : state transition matrix
 - B : measurement transition matrix
 - R : square root of the state noise covariance
 - S : square root of the measurement noise covariance

Gaussian Linear State-Space Models

Advantages:

- Due to **linearity** and **Gaussianity**, this model is one of the very few cases for which exact and reasonably efficient numerical procedures are available to compute the distributions of the X -variables given Y -variables.
- We can use the **Kalman filter** to give closed-form recursive equations for prediction and updating. It is the best possible estimator.

Conditionally Gaussian Linear State-Space Models (CGLSSMs)

More complicated model: Conditionally Gaussian Linear State-Space Models

- The hidden state $X_k = (C_k, W_k)$ includes:
 - A discrete latent process C_k (e.g., regime switching)
 - A continuous Gaussian state process W_k
- Given C_k , the model is a standard linear Gaussian state-space model:

$$\begin{aligned}W_{k+1} &= A(C_{k+1})W_k + R(C_{k+1})U_k \\Y_k &= B(C_k)W_k + S(C_k)V_k\end{aligned}$$

- The indicator process $\{C_k\}$ is a Markov chain with transition kernel Q_C and initial distribution ν_C . Usually C is finite and then identified with the set $\{1, \dots, r\}$.
- U_k, V_k are i.i.d. standard Gaussian noise
- A, B, R, S : known matrix-valued functions of appropriate dimension.

Structure of the Model

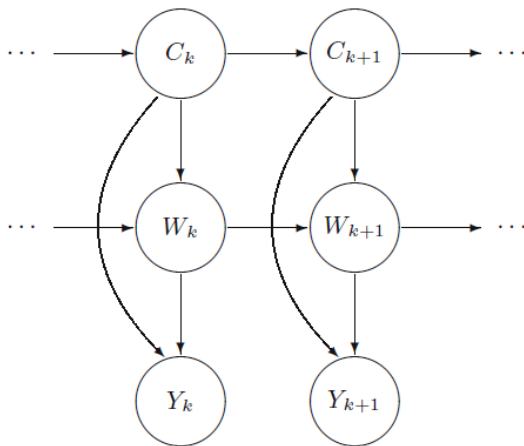


Figure: Graphical representation of the dependence structure of a hierarchical HMM.

Conditionally Gaussian Linear State-Space Models

- The overall system is not Gaussian, but becomes a linear Gaussian state-space model when conditioned on the finite-valued process $\{C_k\}_{k \geq 0}$.
- Conditionally Gaussian models allow us to handle complex, non-Gaussian behavior by stitching together simpler Gaussian models, one per regime. They are especially useful because we can still use efficient tools (like Kalman filters) within each mode.

Example: Rayleigh - Fading Channel

Application: digital communication

Simple Model Construction:

- A signal is **transmitted** and forms an iid sequence of fair Bernoulli draws $\{C_k\}$, taking values in $\{-1, +1\}$
- On the **receiving side**, the signal is demodulated and distorted and may be written as:

$$Y_k = hC_k + V_k$$

- h : channel gain/ fading coefficient
- $\{V_k\}$: iid sequence of Gaussian observation noise with zero mean and variance σ^2

Example: Rayleigh - Fading Channel

Application: digital communication

Realistic Model:

- In a realistic setting, h is time varying, so a very simple model consist in assuming the fading coefficient follows a autoregressive model of first order:

$$W_{k+1} = \rho W_k + U_k$$

$$Y_k = W_k C_k + V_k$$

- Where the time varying h is denoted by W_k .

Goal: Estimate the bits $\{C_k\}$ from the noisy observation $\{Y_k\}$, despite the randomness in the channel W_k . But since the W_k is unknown, that makes the estimations harder.

Inference Challenge

- The model:

$$Y_k = W_k C_k + V_k, \quad \text{with } C_k \in \{-1, +1\}$$

leads to a difficult inference problem:

- If we **condition on a bit sequence** $\{C_k = c_k\}$, the model becomes **linear and Gaussian** in W_k — Kalman filter applies.
- But marginalizing over all possible $C_k \in \{-1, +1\}$ gives a **mixture of 2^k Gaussians**.
- As time progresses, the number of Gaussian components grows exponentially. This makes exact inference intractable.
- However, conditioning on each possible bit sequence $\{C_0, \dots, C_k\}$ gives a tractable linear Gaussian model, and the Kalman filter applies.

- **The Kalman filter; a key tool:**

- We can run **parallel Kalman filters**, each conditioned on a different bit sequence.
- The result is a **mixture of Kalman filters**, each tracking a hypothesis.
- This becomes **computationally heavy**, so other approximations are used in practice.

- **Other methods to handle this complexity:**

- **Gaussian sum filters**: approximate the full mixture with fewer components.
- **Particle filters**: sample and weigh candidate trajectories.
- **Viterbi decoding / EM algorithm**: used when bits C_k are structured (e.g., in communication).

What is the Kalman Filter?

The Kalman filter is a recursive algorithm used to estimate the hidden state of a linear dynamical system from noisy observations. It is optimal under the assumptions of linearity and Gaussian noise.

It is composed of two essential parts:

- **Prediction step:** Uses the previous estimate and system model to predict the state at the next time step, and how uncertain we are about it.
- **Update step (Filtering):** Incorporates the new observation to correct the prediction and reduce uncertainty.

Kalman Filter Algorithm

Algorithm: Full Kalman Filter (Prediction + Filtering)

Initialization:

$$\hat{X}_{0|0} = \mathbb{E}[X_0] = 0 \quad (\text{most common}), \quad \Sigma_{0|0} = \text{Cov}(X_0)$$

These initialization parameters of the model are often picked or given.

Recursion: For $k = 1, \dots, n$, **do:**

❶ Prediction step:

$$\epsilon_k = Y_k - B_k \hat{X}_{k|k-1} \quad \text{innovation / pred. err.}$$

$$\Gamma_k = B_k \Sigma_{k|k-1} B_k^T + S_k S_k^T \quad \text{innovation cov.}$$

$$H_k = A_k \Sigma_{k|k-1} B_k^T \Gamma_k^{-1} \quad \text{Kalman Gain}$$

$$\hat{X}_{k+1|k} = A_k \hat{X}_{k|k-1} + H_k \epsilon_k \quad \text{pred. state estim.}$$

$$\Sigma_{k+1|k} = (A_k - H_k B_k) \Sigma_{k|k-1} (A_k - H_k B_k)^T + R_k R_k^T \quad \text{pred. err. cov.}$$

- ② **Filtering step:** The Kalman Filtering algorithm initial starts with:

$$\begin{aligned}\hat{X}_{k|k-1} &= A_{k-1}\hat{X}_{k-1|k-1} \\ \Sigma_{k|k-1} &= A_{k-1}\Sigma_{k-1|k-1}A_{k-1}^T + R_{k-1}R_{k-1}^T\end{aligned}$$

However we will use the results $\hat{X}_{k+1|k}$ and $\Sigma_{k+1|k}$ of the prediction step, while making sure to shift the index accordingly and compute:

$\epsilon_k = Y_k - B_k\hat{X}_{k k-1}$	innovation
$\Gamma_k = B_k\Sigma_{k k-1}B_k^T + S_kS_k^T$	innovation cov.
$K_k = \Sigma_{k k-1}B_k^T\Gamma_k^{-1}$	Kalman (filter.) gain
$\hat{X}_{k k} = \hat{X}_{k k-1} + K_k\epsilon_k$	filter. state estim.
$\Sigma_{k k} = \Sigma_{k k-1} - K_kB_k\Sigma_{k k-1}$	filter. error cov.

- ③ We now repeat the process and apply the prediction step.

Application to Example Rayleigh - Fading Channel

We are given a linear-Gaussian state-space model. The Kalman filter provides a systematic method to estimate the hidden state sequence $X_0, X_1, \dots, X_n$ from noisy observations $Y_0, Y_1, \dots, Y_n$.

Starting from an initial prior $(\hat{X}_{0|0}, \Sigma_{0|0})$, we recursively apply the prediction and update steps described above. This produces filtered estimates $\hat{X}_{k|k}$ and associated error covariances $\Sigma_{k|k}$, which represent our best guesses of the system's hidden states given the data available up to time k .

Conclusion

- We introduced Hidden Markov Models (HMMs), with a focus on the Gaussian Linear State-Space Models (GLSSMs).
- These models provide a powerful framework for modeling time-evolving systems with noisy observations, such as in digital communication or tracking applications.
- The Kalman filter was presented as an optimal recursive algorithm for estimating the hidden state in GLSSMs, alternating between a prediction and a filtering step.
- The structure of the Kalman filter allows it to be applied efficiently in practice, using only matrix operations and no sampling or brute-force enumeration.
- **Next step:** I aim to implement the Kalman filter numerically and test it on simulated data, particularly in the context of the Rayleigh-fading communication model presented earlier.